

The Alberta Buck - Notes Mutual Decryptability (DRAFT v0.1)

Closing the Issuer-Binding Gap so Every BUCK Note Yields a Provable Receipt

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A monetary system can promise privacy or it can promise accountability; the Alberta Buck promises both at once, along a single seam. *Bulk* traceability is defeated – no chain-analytics firm can reconstruct who paid whom – while *bilateral* disclosure is preserved: the two parties to any payment always retain the cryptographic means to prove, under compulsion, exactly who the counterparty was. The load-bearing half of that promise is the **non-deniable-receipt invariant**: there must be no path by which someone pays BUCKs to another and keeps cryptographic deniability of the act. Concretely, a recipient who is compelled by a court, and who chooses to comply, must be able to produce a cryptographically sound receipt naming the payer's *decrypted, KYC-bound Identity*.

This document shows that BUCK Notes do not yet satisfy that invariant, defines it formally, and specifies the fix. The shipped spend path verifies the *depositor's* identity (`IdentityRegistry.verifySpendCP`), but the mint path verifies *nothing* about the *issuer's*: the per-leaf `issuerData` is folded into the commitment's `idHash` and never checked – exactly the "flavor-specific witness verification (Schnorr for A1/B1, ElGamal re-randomization well-formedness for A2)" that `mint.circom` marked "deferred to a later revision" and that was never re-introduced. A dishonest or colluding issuer can therefore mint a note that pays a recipient while leaving no recoverable payer Identity. The remedy is to make the *deferred approve handshake* symmetric and fully verified: bind a checked issuer identity at mint (a Schnorr binding for public issuers; a recipient-hiding re-encryption proof for the private A2 issuer), and add the `E_depositor_for_issuer` payload to B-note deposits. Each leg is stated as a theorem with an executable `py_ecc` check, and the circom / EVM formal-verification tooling that can mechanize the soundness arguments is catalogued. (Notes, Proofs)

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1 The Non-Deniable-Receipt Invariant

The Alberta Buck identity layer is built so that *every* BUCK transfer leaves both endpoints holding decryptable proof of the other’s Identity. For a direct **transfer**, the bilateral **approve** handshake establishes this: each party re-encrypts their KYC-certified credential under the other’s key and proves (Chaum-Pedersen) that the re-encryption carries the same Identity point M . BUCK Notes defer and reorganise that handshake through the commitment pool, and the invariant must survive the reorganisation.

1.1 Statement

Definition 1. Let an Identity be the KYC-certified point $M \in \mathbb{G}_1$ bound to a registered account by an issuer’s Pointcheval-Sanders credential and stored (ElGamal-encrypted) as $E_{addr}[X] = (R_X, C_X) = (r_X G, M_X + r_X \cdot pk_X)$, with $pk_X = sk_X G$. A payment moves value v from a payer P to a recipient D .

Definition 2. (Non-deniable receipt.) A protocol satisfies the **non-deniable-receipt invariant** if for every accepted payment there exists a receipt $\mathcal{R}$ and an efficient public predicate RcptVerify such that:

- (**Completeness.**) The recipient D can compute $\mathcal{R}$ from data they hold plus public chain state, and $\text{RcptVerify}(\mathcal{R}) = 1$ names a registered Identity M_P and value v .
- (**Soundness / non-frameability.**) No PPT adversary can produce $\mathcal{R}'$ with $\text{RcptVerify}(\mathcal{R}') = 1$ naming an Identity $M_{P'}$ that did not author the payment.
- (**No deniability.**) No PPT payer can cause an accepted payment to D for which no accepting receipt naming the payer exists – even in collusion with D .

The third clause is the operative one: it is the collusion-resistant form. A self-interested recipient who simply wants a record is protected by completeness alone; the invariant additionally forbids a *willing* pair from constructing an untraceable BUCK payment.

1.2 Why the asymmetry between flavors

Who can be named, and how, depends on what is known at mint time – which is exactly the axis that defines the note flavors:

- **B (bearer).** The depositor is unknown at mint, so the note cannot carry a recipient-targeted ciphertext. The *issuer* must therefore be a **public** Identity: their M is in the clear and the recipient names them directly. A bearer note from an *encrypted* issuer is unnameable and must be impossible to mint.
- **A (addressed).** The issuer knows the recipient at mint and has already re-encrypted material the eventual depositor can open. The issuer may stay *private* (A2) by re-encrypting their *own* Identity under the recipient’s key ($E_{\text{iss-for-rec}}$), which only the recipient can decrypt, or *public* (A1) by attaching it in the clear with a signature.

In all cases the recipient must end up holding a path to a *verified* issuer Identity. The next section shows the shipped code does not provide one.

For the recipient to prove "I received v BUCKs issued by Identity M_P ", the receipt must chain four links, each independently checkable against public chain state:

- Links (a), (b), (d) are already sound in the shipped contracts. (b) holds because `cms []` live in the mint transaction’s `calldata`, which is immutable chain history; the recipient reveals `cm` to anchor it (accepting the `spend`↔`mint` linkage, which is the price of complying). (d) holds because the nullifier is a deterministic function of the opening and is emitted by `Notes.Spent / Notes.SpentA` alongside the ERC-20 `Transfer` from the pool.

The invariant lives or dies on link (c) – and (c) is precisely the step the implementation never enforces.

3 What the Shipped Implementation Binds

The spend side verifies the depositor's Identity; the mint side verifies nothing about the issuer's. This asymmetry is the whole bug.

Leg	Shipped check	Guarantee
Depositor (A-spend)	<code>IdentityRegistry.verifySpendCP(spender, recipient, E_n, pi)</code>	spender provably <i>is</i> the addressed M_{rec}
Depositor (B-spend)	none (knowledge of ρ is the authorization)	spender is <i>some</i> registered recipient
Issuer (any mint)	none – <code>idHash</code> folds <code>issuerData</code> via Poseidon-8, value never checked	none

Concretely, from the source:

- `circuits/spend_a.circom` constrains `idHash === Poseidon8(eNoteR.x, eNoteR.y, eNoteC.x, eNoteC.y, issuerData[0..3])`, where `issuerData` is a *private witness* holding A1 ($m_{iss}, \sigma_R.x, \sigma_R.y, \sigma_s$) or A2 (R_{iss}, C_{iss}) = $E_{iss-for-rec}$. The circuit *hashes it in* but never verifies the signature or the encryption.
- `circuits/spend.circom` (the shipped B-spend) leaves `flavor` unconstrained and binds *no* issuer material at all; `idHash` is opaque.
- `circuits/mint_batch.circom` public inputs are `[oldRoot, newRoot, nextLeafIndex, totalFace, cm]` with private `[flavor, v, rho, idHash, predicate, siblings]`. No issuer key, no signature, no ElGamal well-formedness. `mint.circom` says so in as many words: *"flavor-specific witness verification (Schnorr for A1/B1, ElGamal re-randomization well-formedness for A2) is deferred to a later revision."*
- `src/Notes.sol mint()` records `Minted(issuer = msg.sender, ...)` but ties the minting account to the leaves only through transaction authorship, never cryptographically to the committed `issuerData`.

4 The Gap, Per Flavor

4.1 A2 – addressed, private issuer

The opening carries $E_{\text{iss-for-rec}} = (R_{\text{iss}}, C_{\text{iss}})$, an ElGamal ciphertext the recipient decrypts with sk_{rec} to obtain M_{iss} . It is bound into `idHash` but **not** proven to re-encrypt the issuer’s *registered* credential. A malicious issuer commits a random $(R_{\text{iss}}, C_{\text{iss}})$: the note mints and spends cleanly, the recipient decrypts to a meaningless point, and `link(c)` yields no registered Identity. **The payer keeps full deniability.**

4.2 A1 – addressed, public issuer

The opening carries (m_{iss}, σ) . The Schnorr signature σ is never verified, and nothing forces m_{iss} to equal the *minting* account’s registered M . A malicious issuer can omit a valid σ (no receipt), or commit a *victim’s* m_{iss} and a forged σ (frame), and the note still works.

4.3 B1 – bearer, public issuer

`spend.circum` binds no issuer Identity whatsoever. The only issuer signal is `msg.sender` in the `Minted` event. A receipt is therefore constructible *only if* the issuer minted from a **public** Identity – and the contract cannot enforce that, because `cms[]` are opaque and it cannot tell a bearer leaf from an addressed one. A bearer-spendable note minted from an *encrypted* Identity is unnameable. **The payer keeps full deniability.**

4.4 Summary

In every flavor the issuer Identity is, at best, *committed* but never *verified*; in B1 it is not even committed. The depositor half of the handshake is checked (`verifySpendCP`); the issuer half is missing. The fix is to add it.

5 The Fix: A Symmetric, Verified Handshake

Restore the symmetry the "deferred approve handshake" framing always implied: the *mint* is the issuer's half of approve and must be *verified*, exactly as the *deposit* verifies the depositor's half.

5.1 Issuer half, bound at mint

At mint, the issuer is `msg.sender` – *not* anonymous (they authored the tx). So the issuer's registered credential $E_{\text{addr}}[\text{issuer}]$ may be read as a public input; only the *recipient* (for A2) must stay hidden.

- **Public issuer (A1, B1).** Require `msg.sender` to be `isPublicIdentity`, and verify a Schnorr signature by their registered key pk_{iss} over the batch commitment $H_{\text{batch}} = \text{keccak}(\text{cms})$. Bind $m_{\text{iss}} = M_{\text{iss}}$ (the in-the-clear registered Identity) into `idHash`. Because the registered Identity of a public account is in the clear, link (c) reduces to a registry read plus the verified signature.
- **Private issuer (A2).** The issuer proves `E_iss-for-rec` re-encrypts their *own* registered M_{iss} under the recipient's key pk_{rec} – the `verifyApprove` relation with `sender = issuer`, `spender = recipient`. But `verifyApprove` as shipped takes pk_{rec} as a verifier input and so *names the recipient*, which A2 must not do. Two recipient-hiding realisations:

1. **In-SNARK (mint circuit).** Extend `mint_batch.circom` so each A2 leaf proves, with pk_{rec} and M_{iss} as private witnesses:

$$pk_{\text{iss}} = sk_{\text{iss}}G, \quad M_{\text{iss}} = C_{\text{iss}}^{\text{reg}} - sk_{\text{iss}}R_{\text{iss}}^{\text{reg}}, \quad E_{\text{iss-for-rec}} = (r'G, M_{\text{iss}} + r'pk_{\text{rec}}),$$

with $E_{\text{addr}}[\text{issuer}] = (R_{\text{iss}}^{\text{reg}}, C_{\text{iss}}^{\text{reg}})$ a *public* input and pk_{rec} the same key that re-randomises the note ciphertext E_{note} . Cost: a handful of in-circuit $\mathbb{G}_1$ scalar muls per A2 leaf – the ~10x the original deferral feared, but bounded and paid only on A2 mints.

2. **Recipient-blinded on-chain CP.** Keep verification on-chain (EIP-196, cheap) but blind pk_{rec} : the issuer proves the re-encryption relation against a *Pedersen commitment* to pk_{rec} that also appears (identically committed) in E_{note} , so the verifier never learns pk_{rec} . This is a new sigma protocol (Section *Provable Correctness*), avoiding in-SNARK $\mathbb{G}_1$ arithmetic at the cost of a bespoke proof.

The in-SNARK route (1) is the recommended default: it reuses the trusted setup already in place for the batch mint and keeps the on-chain mint a single verifier call.

5.2 Depositor half, at deposit

- **A1, A2.** Already shipped: `verifySpendCP` proves the spender is the addressed M_{rec} . The issuer, who chose M_{rec} at mint, already holds the symmetric knowledge.
- **B1.** Add `E_depositor_for_issuer` to the spend: the depositor re-encrypts *their* registered M_{dep} under the public issuer's pk_{iss} , with a `verifyApprove`-style CP proof (`sender = depositor`, `spender = issuer`, both public-or-named here so no blinding is needed), emitted in the `SpentB` event. This closes the symmetric direction – the public bearer issuer can identify who cashed their note.

5.3 Enforcing bearer $\Rightarrow$ public issuer

The contract cannot read `flavor` from an opaque `cm`. Make the mint flavor-aware for the binding: the mint circuit exposes, per leaf, a public `issuerMode` in `{PUBLIC, PRIVATE}` and proves the matching binding (Schnorr-bound public M , or in-SNARK re-encryption). `Notes.mint` requires every `PRIVATE` leaf to be an A-flavor (addressed) leaf – a bearer (B) leaf with `issuerMode = PRIVATE` is unsatisfiable – and requires `msg.sender` to be `isPublicIdentity` whenever any `PUBLIC` leaf is present.

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6 Protocol Changes (Implementation Surface)

6.1 IdentityRegistry

```
// Public-issuer binding: Schnorr by msg.sender's registered key over H_batch.
function verifyIssuerSchnorr(address issuer, bytes32 hBatch, SchnorrProof calldata s)
    external view returns (bool);

// Private-issuer binding (A2), recipient-blinded sigma variant (option 2):
// proves E_iss_for_rec re-encrypts E_addr[issuer]'s M under a *committed* pk_rec
// that the mint SNARK also binds to E_note, without revealing pk_rec.
function verifyIssuerCPBlinded(address issuer, ElGamalCT calldata eIssForRec,
    bytes32 pkRecCommit, IssuerCPPProof calldata pi)
    external view returns (bool);

// B-note depositor->issuer (mirror of verifyApprove, both parties named):
function verifyDepositorForIssuer(address depositor, address issuer,
    ElGamalCT calldata eDepForIss, CPPProof calldata pi)
    external view returns (bool);
```

Under the in-SNARK route the A2 binding is discharged inside the mint verifier and `verifyIssuerCPBlinded` is unnecessary; the registry then only gains `verifyIssuerSchnorr` and `verifyDepositorForIssuer`.

6.2 Notes.sol

- `mint(...)` gains per-leaf `issuerMode[]` (public input of the mint SNARK) and, for public leaves, a `SchnorrProof` over `keccak(cms)`; it requires `msg.sender` public when any public leaf is present, and rejects `PRIVATE+bearer` leaves. The A2 re-encryption binding rides inside the mint proof (option 1) or is checked per leaf via `verifyIssuerCPBlinded` (option 2).
- `spend(...)` (B) gains `(E_depositor_for_issuer, CPPProof)`, calls `verifyDepositorForIssuer(msg.sender, issuer, ...)`, and emits `SpentB(nullifier, face, recipient, eDepForIss)`. The issuer address is recovered from the leaf's public-issuer binding revealed at `spend`.
- `spendACP(...)` (A) is unchanged on the depositor side; the issuer guarantee now comes for free from the mint-time binding.

6.3 Circuits

- `mint_batch.circom`: add the per-leaf `issuerMode` public signal and, for A2 leaves, the re-encryption constraints above (option 1). For public leaves, constrain `idHash`'s `issuerData` prefix to equal the registered public `M` so the in-clear identity is the one bound.
- `spend.circom` (B): constrain `flavor == B1=` and reveal enough of the leaf's public-issuer binding to recover the issuer address on-chain.
- `spend_a.circom`: unchanged (the A2 issuer binding moved to mint).

6.4 Wallet (`alberta_buck.wallet`)

- Mint: construct the issuer Schnorr proof (public) or the A2 re-encryption witness (private), mirroring `notes.id_hash_a1/a2`.
- Spend (B): construct `E_depositor_for_issuer` and its CP proof.
- Receipt: implement `RcptVerify` as a standalone verifier (opening + mint-tx reference + spend-tx reference + issuer recovery) so a recipient can produce and a third party can check the receipt entirely off-chain from chain data.

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7 Provable Correctness

We extend the companion *Proofs* document (Theorems 1-11) with the issuer-binding results. Each is accompanied – in the proofs document’s house style – by an executable `py_ecc` sanity check on BN254.

7.1 Theorems

Theorem 3. (Issuer-binding completeness.) *An honestly generated public-issuer Schnorr proof, and an honestly generated A2 re-encryption proof (in-SNARK or blinded), verify.*

Theorem 4. (Issuer-binding soundness.) *If `verifyIssuerSchnorr` accepts, the batch commitments were authorised by `issuer`’s registered key. If the A2 re-encryption proof accepts, $E_{iss-for-rec}$ decrypts under sk_{rec} to the issuer’s registered M_{iss} (the same M certified by the issuer’s PS credential – Theorem 6). Reduction: special soundness of the underlying sigma protocol (two accepting transcripts sharing commitments extract sk_{iss}, r'), as in Theorems 4, 6; or Groth16 knowledge-soundness for the in-SNARK route.*

Theorem 5. (Receipt completeness.) *For every note accepted by the patched `mint` and redeemed by `spend` $\neq$ `spendACP`, the recipient can compute $\mathcal{R}$ with $RcptVerify(\mathcal{R}) = 1$ naming a registered Identity. Proof: links (a),(b),(d) hold as before; (c) now holds by issuer-binding soundness (public: registry $M + Schnorr$; private: decrypt $E_{iss-for-rec}$, validity guaranteed by the accepted binding).*

Theorem 6. (Receipt soundness / non-frameability.) *No PPT adversary produces an accepting $\mathcal{R}'$ naming $M_{P'}$ that did not author the payment. Reduction: a forged public receipt breaks Schnorr EUF-CMA; a forged private receipt breaks re-encryption soundness (Theorem 7) or the binding hash (collision resistance of Poseidon-8 in `idHash`).*

Theorem 7. (No deniability under collusion.) *No payer-recipient coalition can produce an accepted payment with no accepting receipt: `mint` rejects any leaf lacking a valid issuer binding, so every spendable leaf carries one, and the binding is recoverable by the holder of the opening.*

Theorem 8. (Recipient privacy preserved.) *The A2 issuer binding is zero-knowledge in pk_{rec} and M_{iss} : the in-SNARK route reveals only the Groth16 proof; the blinded route reveals only a hiding Pedersen commitment to pk_{rec} . No observer learns the recipient. Reduction: HVZK of the sigma protocol (Theorem 5) / Groth16 zero-knowledge.*

Corollary 9. (Mutual decryptability.) *Combining the issuer half (above) with the depositor half (`verifySpendCP` for A; `verifyDepositorForIssuer` for B), both parties can recover the other’s registered Identity, and each can prove it under compulsion. This is the Notes specialisation of the BUCK mutual-decryptability invariant (Proofs Theorem 11), now enforced rather than assumed.*

7.2 Executable sanity checks (py_ecc on BN254)

Mirroring the proofs document, each theorem ships a small witnessed check that runs in CI alongside the existing `alberta_buck.wallet` vector generation:

- **Issuer Schnorr:** sign H_{batch} under sk_{iss} , assert verify; assert a tampered H_{batch} or wrong key fails.

- **A2 re-encryption:** build $E_{\text{addr}}[\text{issuer}]$, decrypt to M_{iss} , form $E_{\text{iss-for-rec}}$ under pk_{rec} , run the prover/verifier; assert the recipient's sk_{rec} -decryption equals M_{iss} ; assert a random $E_{\text{iss-for-rec}}$ is rejected (soundness witness) and that the transcript hides pk_{rec} (two runs with distinct pk_{rec} are indistinguishable to the verifier given the commitment).
- **Receipt:** assemble $\mathcal{R}$ from a synthetic mint+spend and assert $\text{RcptVerify} = 1$; assert a frame attempt (swap M_P) fails.

These vectors double as the canonical inputs the Solidity and circom tests load via `vm.readFile`, keeping the on-chain verifier, the circuit, and the reference math in lock-step (the pattern already used for `verifySpendCP`).

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8 Formal-Verification Approaches

"Every step provably correct" spans three artifact classes – the sigma-protocol mathematics, the circom circuits, and the Solidity verifiers – each with its own tractable tooling.

8.1 The cryptographic mathematics

- **Hand proofs** + `py_ecc`: the existing *Proofs* methodology – completeness, special-soundness (2-extractability), and HVZK arguments, each pinned by an executable BN254 check. Low-tech, high-confidence, already in the repo.
- **EasyCrypt**: a proof assistant purpose-built for *game-based* cryptographic proofs; the sigma-protocol soundness and ZK lemmas can be mechanised as reductions, giving machine-checked versions of Theorems above.
- **CryptoVerif** (computational) or **ProVerif** (symbolic): protocol-level analysis of the full handshake (mint binding + deposit binding) against the non-deniability and recipient-privacy goals stated as queries.

8.2 The circom circuits (soundness is the hard part)

The danger in arithmetic circuits is *under-constraint* – a witness slot that the prover can set freely, silently widening soundness (the exact failure mode of the unverified `issuerData`). Tooling that targets this directly:

- **Picus** (Veridise): automated detection of under-constrained signals; run it on the patched `mint_batch.circom` to prove the new issuer-binding signals are *determined* by the public inputs.
- **Circomspect** (Trail of Bits): static analysis / linting for common circom pitfalls (unconstrained signals, `<--` without `<=`, `=` signal-vs-constraint confusion).
- **ecne** (0xPARC): proves R1CS systems compute deterministic functions of their inputs – a soundness witness for the binding constraints.
- **Coda** / **circom-to-Coq** efforts: full functional-correctness extraction to a proof assistant for the security-critical templates (the re-encryption gate).
- Discipline: keep the security-critical constraints (the $\mathbb{G}_1$ relations) in small, separately-audited sub-templates with explicit `Num2Bits` range checks, as the value-conservation bound already does.

8.3 The Solidity verifiers and Notes.sol

- **Halmos** (a16z): symbolic-execution property tests in Solidity – assert, for all inputs, e.g. "mint reverts unless every leaf carries a valid issuer binding" and "spend emits a recoverable issuer for every accepted note."
- **Kontrol** / **KEVM** (Runtime Verification): K-framework formal proofs over EVM bytecode for the highest-assurance properties (the binding-or-revert invariant, nullifier non-reuse).

- **Certora Prover:** CVL rules for state invariants (`noteFaceSum` conservation, "no accepted spend without a prior valid mint binding").
- **solc SMTChecker:** lightweight, in-compiler assertions and overflow checks on the new code paths.
- **Foundry:** fuzz + stateful invariant testing as the always-on backstop; negative tests below are expressed here first.

8.4 What "provably correct per step" buys

The chain of guarantees is: `=py_ecc=`/EasyCrypt prove the *math*; Picus/ecne prove the *circuit* faithfully and exhaustively enforces that *math*; Halmos/Kontrol prove the *contract* admits no spend whose mint lacked the binding. Composed, no accepted Note payment can exist without a recoverable, sound receipt – the invariant, mechanically checked end-to-end.

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9 Test Plan

9.1 Positive

- A1/A2/B1 honest mint + spend; recipient runs `RcptVerify` and a third party re-checks it – all return 1 naming the correct registered Identity.
- A2 recipient privacy: two mints to distinct recipients are observationally indistinguishable on chain.

9.2 Negative (these must now fail, and did not before)

- A2 issuer commits a random `E_iss-for-rec`: `mint reverts` (binding fails).
- A1 issuer commits a victim's m_{iss} / a forged σ : `mint reverts`.
- Bearer leaf with `issuerMode = PRIVATE`, or any public leaf from a non-`isPublicIdentity` minter: `mint reverts`.
- Recipient attempts a forged receipt naming a non-issuer: `RcptVerify = 0`.

9.3 Cross-artifact

- The `py_ecc` vectors drive both the circom witness tests and the Solidity verifier tests, asserting byte-identical `idHash` / proof encodings (the established `alberta_buck.wallet -> vm.readFile` pattern).
- Picus run in CI gates merges on "no new under-constrained signals."

10 Phasing and Migration

1. **Public issuers first (A1, B1).** Schnorr binding + `isPublicIdentity` enforcement + `verifyDepositorForIssuer` for B. No new circuit setup; pure contract + wallet work. Closes the gap for the public-issuer flavors, which are the audit-sensitive payroll / cashier-cheque cases.
2. **Private issuer (A2).** The in-SNARK re-encryption binding: extend `mint_batch.circom`, re-generate the per-N verifiers and trusted setup, wire `issuerMode`. The larger lift; ship behind a governance-set verifier slot so the new mint verifier can be rotated in (`Notes.setMintVerifier`).
3. **Mechanisation.** Land Picus/Halmos in CI; mechanise the sigma proofs in EasyCrypt as time allows.

Notes minted under the old (unbound) verifier remain spendable but do not carry the guarantee; governance announces a cutover root after which only bound mints are accepted, and wallets surface a `UNVERIFIED ISSUER` status for pre-cutover notes (a new wallet status alongside `VALID` `!=DEAD` `!=FAKE`).

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11 Implementation Status (Phase 1 shipped)

This section records what is **actually shipped** against the spec above, so the state is self-contained. Work proceeds on branch `feature/decryptability` (cut from `master` after the `feature/buck-negative-balance` merge, `f445905`).

11.1 Phase 1 – shipped, wired, and tested

The public-issuer half and the bearer depositor->issuer half are implemented end-to-end. Commits, in order:

- `d8dc818` – `IdentityRegistry.verifyIssuerSchnorr + SchnorrProof` struct + `_fsIssuerSchnorr` helper. Checks $s * G = R + e * pk_{iss}$, gated on `isPublicIdentity[issuer]`; Fiat-Shamir binds $(pk_{iss}, R, hBatch, issuer, chainid)$.
- `0df3e71` – `Notes.mint` split into a 6-arg (legacy) and 7-arg (+ `SchnorrProof`) overload over an internal `_mint`. When `isPublicIdentity(msg.sender)`, `_mint` **requires** `verifyIssuerSchnorr(msg.sender, keccak256(cms), sig)`; the 6-arg path passes a zero proof and so reverts for a public issuer. New `IssuerBound(issuer, newRoot)` event.
- `5f92cdc` – wallet `alberta_buck/wallet/schnorr.py` (`issuer_schnorr_sign/verify, batch_commitment = keccak of packed cms`); `vectors.py` emits an `issuer_schnorr` section into `test/vectors/identity.json`; the FS transcript matches `_fsIssuerSchnorr` byte-for-byte.
- `0f0ab0f` – `IdentityRegistry.verifyDepositorForIssuer` (extracted the `verifyApprove` body into `_verifyApprove`; the alias just relabels sender/spender as depositor/issuer). `Notes.spend` split into a 5-arg (legacy) and 8-arg (+ `issuer, eDepForIss, CPProof`) overload over `_spend`; the 8-arg path checks `verifyDepositorForIssuer(recipient, issuer, ...)` and emits `SpentB`.
- (working tree) – `RcptVerify`, the off-chain receipt verifier (`alberta_buck/wallet/receipt.py`): a `Receipt` (opening + minted batch + issuer Schnorr + Spent-event nullifier/face) and `receipt_verify` that checks the four-link chain for **public-issuer** notes (A1, B1) and names the issuer's registered Identity, refusing A2 (private issuer, Phase 2). No secret is needed to verify, so a third party can re-check it. A canonical `receipt` section was added to `test/vectors/identity.json` (Bob, a public Corporate Identity, mints a B1 bearer note Alice cashes). This is the Phase 1 capstone: links (a),(b),(d) were already sound on-chain and (c) is now bound for public issuers, so the receipt is finally **produceable**.
- (working tree) – the symmetric **direct EOA-transfer** receipt. A new `alberta_buck/wallet/verifiable_` primitive (a Chaum-Pedersen DLEQ on bases (G, R) , the `spend_cp` shape specialised to $(R, C-M)$) proves an ElGamal ciphertext decrypts to a **revealed** M under a registered key without exposing sk – the "no deniability under collusion" clause for an encrypted-identity counterparty. `receipt.py` gains `ApproveReceipt + approve_receipt_verify`, composing the sender's approve handshake (`chaum_pedersen_verify / verifyApprove, soundness`) with the recipient's verifiable decryption (recovery + provability) to name the sender's registered Identity, third-party-checkable with no secret. This packages what the wallet previously left a la carte (`chaum_pedersen + elgamal_decrypt`) and shares `RegisteredIdentity / RcptResult` with the Notes verifier. An `approve_receipt` section (Bob names Alice from the Alice->Bob approve) is added to `identity.json`. No on-chain verifier ships (off-chain, like `RcptVerify`);

the verifiable-decryption primitive is also the building block the A2 Notes receipt will reuse (decrypt `E_iss-for-rec`, prove correctness).

Tests – all green; full forge suite **383 passed, 1 skipped** (the offline-skipped `BuckKControllerForkTest`; the receipts are wallet-only so the forge count is unchanged; the wallet pytest set is **131 passed**):

- `test/IssuerSchnorr.t.sol` – 7 in-Solidity (completeness + soundness: tampered batch / response, wrong key, non-public, unregistered, cross-issuer replay) + 4 `IssuerSchnorrVectorTest` (the python-emitted proof verifies on-chain – the cross-artifact parity check).
- `test/NotesIssuerBinding.t.sol` – 5 (valid binding mints; tampered / wrong-batch sig and the 6-arg overload revert `Notes: bad issuer binding`; non-public minter skips).
- `test/NotesDepositorBinding.t.sol` – 5 (registry parity reusing the `approve` vector as a depositor->issuer proof; bound spend emits `SpentB`; bad CP reverts `Notes: bad depositor binding`; legacy 5-arg unbound works).
- `alberta_buck/test/test_schnorr.py` – 6 (round-trip, tamper, keccak parity).
- `alberta_buck/test/test_receipt.py` – 17 (Notes: completeness names Bob's registered M; third-party recheck needs no secret; soundness: tampered opening, unregistered / non-public / wrong-key issuer, forged batch sig, mismatched nullifier or face all reject; A2 refused. EOA ApproveReceipt: completeness names Alice; third-party recheck; frame-wrong-M, tampered cp/vd proof, substituted sender record, unregistered sender all reject).
- `alberta_buck/test/test_verifiable_decrypt.py` – 6 (round-trip; wrong M, wrong key, tampered response, replayed account/chain reject; the real key still cannot name a false M).

11.2 Decisions / deviations from the spec text above

- **The Schnorr binding is at mint, gated only on the minter being public.** There is NO per-leaf `issuerMode` public signal yet, so the "Enforcing bearer > public issuer" subsection is **not** implemented: Phase 1 binds **public minters only**; the contract still cannot tell a bearer leaf from an addressed one (=cms opaque).
- **`verifyDepositorForIssuer` reuses the `verifyApprove` relation verbatim** – no new sigma protocol, no new wallet signer, no new vector. The existing `approve` (Alice->Bob) vector **is** a valid depositor(Alice)->issuer(Bob) proof. FS binds (depositor, issuer, chainid); the specific note is tied in only by `SpentB` emitting (nullifier, `eDepForIss`) atomically – the nullifier is NOT in the CP transcript.
- **`Notes.spend` binds the recipient (the beneficiary), not `msg.sender`.**
- **Both bindings are optional overloads** (non-breaking): the 6-arg mint and 5-arg `spend` still serve legacy / non-public paths.
- The shipped event is `SpentB(nullifier, face, recipient, issuer, eDepForIss)` (issuer included). The depositor **names issuer** as a parameter; it is **not** recovered from the leaf, because the B-spend circuit does not reveal it.

11.3 Residual gaps -> Phase 2 (NOT done)

1. **Bearer-from-non-public is still unbound** – needs the mint SNARK to expose a verified per-leaf `issuerMode` so `Notes.mint` can reject a bearer leaf from a non-public minter.
2. **The depositor names the issuer at B-spend** (not cryptographically tied to the note) – sound only for an honest depositor; binding the issuer into the B-spend circuit removes this.
3. **A2 private-issuer in-SNARK re-encryption binding** – the central Phase 2 item; extend `mint_batch.circom` + regenerate per-N verifiers / trusted setup, behind `Notes.setMintVerifier`.
4. **RcptVerify for A2** – the public-issuer (A1/B1) verifier and the direct EOA-transfer `ApproveReceipt` are shipped (working tree); the A2 Notes path is refused pending the gap #3 in-SNARK binding that gives it a verified issuer to name. The `verifiable_decrypt` primitive needed to **reveal** the decrypted A2 issuer (`E_iss-for-rec`) now exists, so the A2 receipt reduces to wiring that binding plus reusing the primitive. An on-chain `RcptVerify` (so a contract, not just an off-chain reader, can gate on a receipt) is out of scope here – Phase 1 keeps verification off-chain from chain data.
5. **Formal verification** (Picus / Halmos / EasyCrypt) – not started.

11.4 Cross-references and house conventions

- `alberta-buck-notes.org` was updated this session: the "Mutual Decryptability" invariant gained an "assumed vs. enforced" note, the "Note Flavors" intro gained a "why bearer forces a public issuer – and how it is enforced" paragraph, and Flavor B1 gained a Phase-1 status note. That `.org` edit and the regenerated PDF/TXT exports are **uncommitted**; the Phase 1 code/tests above are committed.
- Conventions in force: a negative signed BUCK balance is **not** "debt" – it is "used credit" / an outstanding claim on the holder's own assets; the carrying transfer is the `isCarrying-dispatched Buck._carryingTransfer (buckSeconds apportionment)`, **not** a `transferCarrying` primitive. Build and test via `make nix-test`; Python via `nix develop --command python -m pytest`. Commit messages: ASCII only, no co-author/AI attribution.

12 Open Questions

1. **In-SNARK vs blinded-CP for A2.** The in-SNARK route is simpler to reason about (one verifier) but enlarges the trusted setup; the blinded-CP route keeps the SNARK small but adds a bespoke sigma protocol to audit. The recommendation is in-SNARK, but the blinded-CP construction is worth a full soundness/ZK write-up before deciding.
2. **Batch vs per-leaf issuer signature.** A single Schnorr over `keccak(cms)` is cheapest but binds the *whole batch* to one public issuer; mixed-issuer batches need per-leaf signatures or per-issuer sub-batches.
3. **Receipt non-repudiation scope.** `RcptVerify` proves the issuer *authored* the note; tying that to a real-world legal identity still rests on the KYC binding of *M* (the PS credential). This document assumes that binding; it does not re-derive it.
4. **Predicate-bearing notes.** The `predicate` field is out of scope here; a programmable predicate that itself moves value would need its own receipt analysis.

DRAFT